

Assignment #6: Database Design

Description:

The goal is to learn how to build the Data Model for a shopping cart website you will create in assignment 7. Your Data Model will consist of 5 entities that store information about the customers, the products and the purchases. The entities will be related according to the Line-Item Pattern discussed in Chapter 5. You will create a Data Model diagram showing how the entities are related using MS Visio 2013 or 2016. Your diagram should use Crow's Foot Notation. The following are details describing the five entities of your database:

	Column Name	Data Type	Allow Nulls
	ID	int	☐
	Firstname	nvarchar(30)	☐
	Lastname	nvarchar(30)	☐
	StreetAddress	nvarchar(50)	☒
	City	nvarchar(30)	☒
	State	nvarchar(20)	☒
	Zip	nvarchar(10)	☒
	Phone	nvarchar(15)	☒
	Email	nvarchar(30)	☐

The Customers Entity

	Column Name	Data Type	Allow Nulls
	ProductID	int	☐
	ProductType	nvarchar(30)	☐
	ProductName	nvarchar(50)	☒
	ImageFile	nvarchar(30)	☒
	UnitPrice	decimal(18, 2)	☐
	MaxAmount	int	☐
	DefaultAmount	int	☐

The Products Entity

	Column Name	Data Type	Allow Nulls
	OptionID	int	☐
	OptionType	nvarchar(30)	☐
	OptionName	nvarchar(50)	☐
	OptionCost	decimal(18, 2)	☐

The Options Entity

	Column Name	Data Type	Allow Nulls
	OrderID	int	☐
	CustomerID	int	☐
	Date	datetime	☐
	SubTotal	decimal(18, 2)	☐
	Tax	decimal(18, 2)	☐
	Total	decimal(18, 2)	☐

The PurchaseOrders Entity

	Column Name	Data Type	Allow Nulls
	OrderID	int	☐
	LineNumber	int	☐
	ProductID	int	☐
	OptionID	int	☒
	Quantity	int	☐
	ExtendedPrice	decimal(18, 2)	☐

The LineItems Entity

Submission Criteria:

1. Turn in a Visio Diagram document called ShoppingCart.vsd with your Data Model in it that uses the Line-Item Pattern from Chapter 5.
2. Submit your assignment through D2L in the assignment area labeled assignment 6.